

Adi Mukundan

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EDUCATION

University of Iowa

Master of Computer Science

Iowa City, IA

Dec. 2026 expected

Relevant Coursework: Computer Security, Intro to Networks, Math of ML, Optimization Techniques

University of Iowa

Bachelor of Science in Computer Science and Mathematics

Iowa City, IA

August 2025

Artificial Intelligence, Modeling and Simulation (AIMS) Certificate, University Honors

EXPERIENCE

NLP Researcher - UIowa Biocomputing Research Lab | *Word2Vec, Ollama* Jan 2025 – Present

- Built RCPD lexicon expansion pipeline (Word2Vec + LLM filtering), identifying new terms related to the disease.
- Recreated WordNet synsets for RCPD lexicon using structured prompts with LLMs, giving a domain tuned vocab.
- Using expanded RCPD vocabulary and interactive embedding visuals to make clinically actionable insights.
- Applied PCA and KMeans to the expanded vocabulary to cluster posts into clinically meaningful categories.

NSF Researcher - Singh Lab | *LLMs, MongoDB, Data Visualization* May 2024 - Oct. 2024

- Developed LLM coding pipeline to label social media themes for epidemiological analyses of opioid use.
- Managed and queried a MongoDB database of 80M+ Reddit posts; transformed slang into actionable visuals.
- Supported a study on deep learning for schistosomiasis treatment measurement by providing image segmentations.

SWE Intern - Siemens | *pytest, Tkinter, Simcenter* May 2023 – Jan 2024

- Created a Python application for development and testing of Simcenter by Motion and Solver Teams.
- Application used by over 10 engineers across 3 teams, saving hours per engineer during development.
- Maintained the codebase and streamlined install/docs with Mercurial; containerized the stack with Docker.
- Streamlined the development process for the Simcenter development teams by simplifying data visualization

SWE Intern - Driving Safety Research Institute | *OpenCV, TensorFlow* Oct. 2022 – May. 2023

- Applied principles of computer vision to real time video feeds with navigation.
- Developed, debugged, and refactored Python code to run on a Raspberry Pi.
- Trained and developed a deep learning model capable of navigation using TensorFlow and Google Colab.
- Reduced image capture and processing time by over 0.6 seconds by leveraging knowledge of Picamera API.

Events Chair - Association for Computing Machinery Spring 2023 - 2024

- Planning annual University of Iowa Computing Conference during Spring semester.
- Setup and run weekly meetings for the University of Iowa chapter of ACM.
- Communicating with professors within the CS department to align faculty and executive board goals for the club.

PROJECTS

Drug Use Insights | *Django, PostgreSQL, Bokeh* | <https://dupont.cs.uiowa.edu/dui/> Spring 2025

- Refactored and deployed a Django webserver on a Debian environment
- Visualizes text analysis of drug use on social media using Bokeh
- Maintains a PostgreSQL database for storage of user-uploaded data

Lifeguard VR Simulation | *Unity, C#* 2024 - 2025

- Optimized graphics and I/O pipeline to ensure above 90fps performance of a continuous search task
- Developed an immersive VR pool environment in Unity to study lifeguard rescue behavior
- Presented findings and demoed environment at the University of Iowa SURF 2025

HONORS & AWARDS

Jason and Leslie Weber Emerging Technologies Scholarship (2025) | John and Elsie Mae Ferentz Research Fellowship (2024) | Computer Science REU (2024) | Iowa Flagship Award (2021 - 2024)

TECHNICAL SKILLS

Languages: Python, Java, C#, SQL, R, JavaScript, C/C++, MATLAB

AI/ML/CV: PyTorch, TensorFlow, scikit-learn, spaCy, Hugging Face, gensim, OpenCV, Plotly, pandas, NumPy

Data/DevOps: MongoDB, PostgreSQL, Docker/Compose, Git/Mercurial, CI/CD, ROS, Streamlit